



Service Bulletin: Errors 33,1010 & 16 – Drive Wheel Malfunction

Product: All Roomba Combo 2025 & 2026

Issue: Errors 33, 1010 & 16 – Drive Wheel Failure

1. Affected Modes of Operation

Error 33, Error 1010 & Error 16 may occur during:

- **Cleaning cycle (Auto / Scheduled cleaning)**
- **Docking / returning to base**
- **Manual control via App or buttons**

The error is triggered when **drive movement is compromised**.

2. Customer Reported Symptoms

Customers may report one or more of the following:

- Robot displays **Error 33 (Navigation), Error 1010 (docking), Error 16 (Robot Lifted)**
- Robot **fails to move correctly**
- Robot **moves in circles or arcs**
- One side of the robot **is not driving / dragging**
- Robot appears **stuck despite clear surroundings**
- Robot indicates **it was lifted off the ground, without any user interaction**
- Robot has **issues driving on and off the dock**

The condition is typically linked to **one non-functioning drive wheel**

3. Important Diagnostic Note (Intermittent Behaviour)

- The issue **may be intermittent** and **not consistently reproducible** during workshop testing
- The robot may pass **mission testing without triggering any Errors**
- Despite no fault observed during testing, the **underlying drive wheel failure may still be present**

Recommendation:

If Error 33, Error 1010 or Error 16 is reported or suspected:

- Perform **precautionary replacement of both left and right drive wheel modules**, even if only one side appears faulty

This reduces repeat failure risk and avoids returns due to **intermittent recurrence**.

4. Initial Troubleshooting (Call Centre Guidance)

Before arranging repair, confirm:

1. **Check for environmental obstruction**
 - Ensure robot is not trapped or blocked
 - Remove any debris or obstacles around wheels
2. **Confirm wheel behaviour**
 - Check if **one wheel is not rotating**
 - Observe:
 - Circular movement
 - Uneven driving

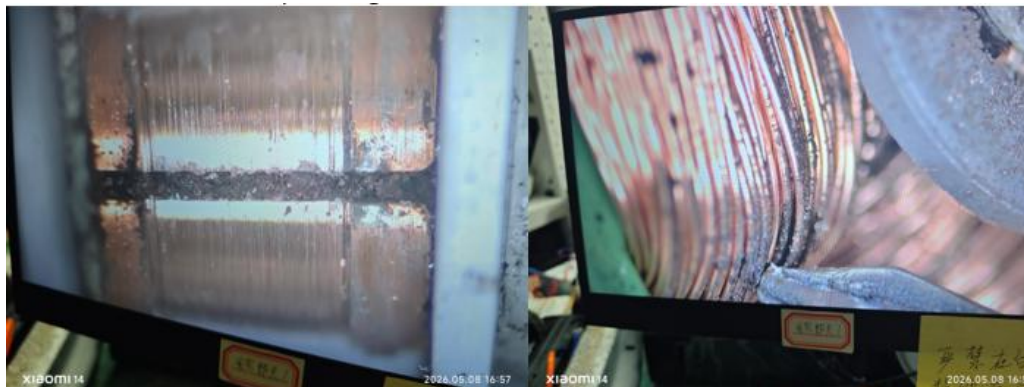
If obstruction is cleared and issue remains → **Proceed to repair**

5. Root Cause (Engineering Analysis)

Analysis of returned units identified:

- **Grease contamination (KE grease)** entering the motor
- Resulting in:
 - **Carbon dust buildup** on coils and commutator
 - **Electrical short circuit within the motor**

This leads to **drive wheel failure and Error 33 trigger**



6. Repair Action (Service Centres)

When Error 33, Error 1010 or Error 16 is confirmed:

- **Standard corrective action:**
 - Replace the **affected drive wheel module**
- **Recommended best practice:**
 - Replace **both drive wheel modules** where Error 33, Error 1010 or Error 16 is reported or suspected (due to intermittent nature)
- Do NOT attempt:
 - Partial repair of motor internals
 - Cleaning of internal motor components

Replacement is required for full resolution

7. Engineering Improvement (In Progress)

- Current mitigation under validation:
 - Replacement of **KE grease** → **FN grease**
 - Objective:
 - Prevent grease migration into motor
 - Reduce recurrence risk
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8. Summary

- Error 33, Error 1010 or Error 16 indicates **drive wheel failure (typically one side)**
- Fault may be **intermittent and not visible during testing**
- Root cause linked to **motor contamination and electrical shorting**
- **Precautionary replacement of both wheels is recommended**
- Customer troubleshooting should rule out simple obstructions first